

An Industry Standard for Software to Hardware Design Flow Integration

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Structure of this Presentation

- Benefits of The SPIRIT Consortium Standards to ARM
- Status of The SPIRIT Consortium
- SPIRIT IP-XACT for ESL to RTL Flow Integration
- SPIRIT IP-XACT for Debug Configuration
- Demonstration of IP-XACT Flow Integration
- In Summary

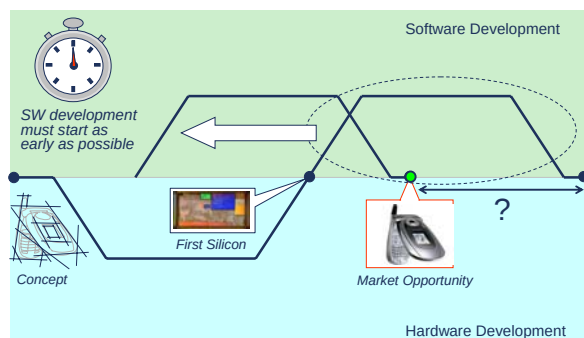


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How Can You Meet Market Demands?

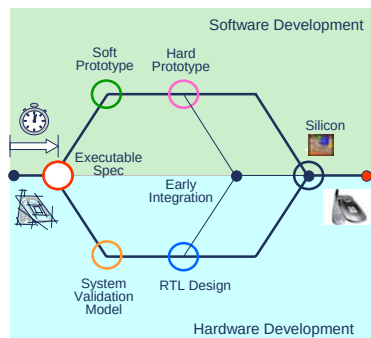


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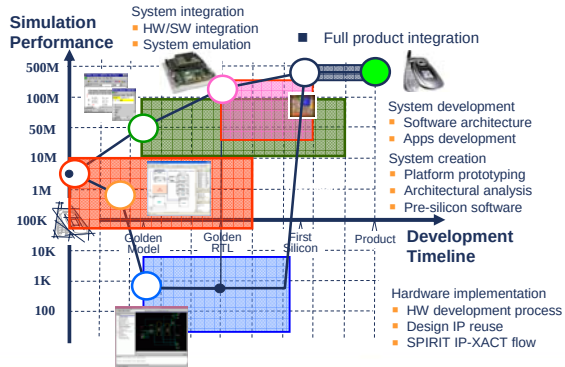


Entering Concurrent Development

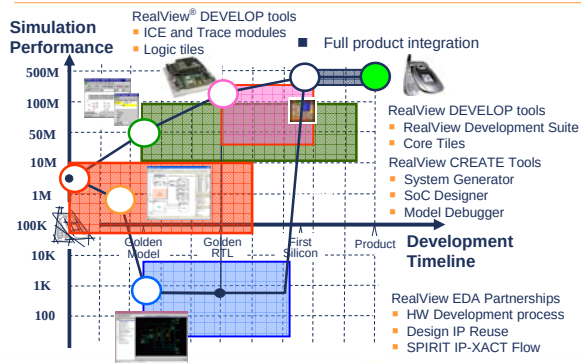


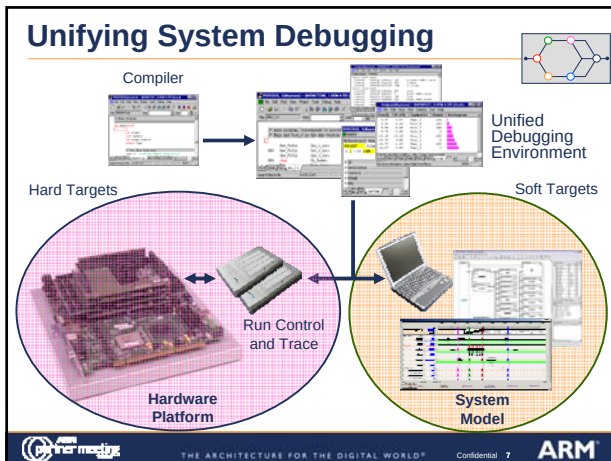
- Key theme:
 - Keep creating executable targets
- Target types:
 - System exploration model (mixed accuracy)
 - System validation model (cycle accurate)
 - Soft prototype (fast cycle approx.)
 - Hard prototype (hard platform with FPGA config.)
 - System test board (silicon)

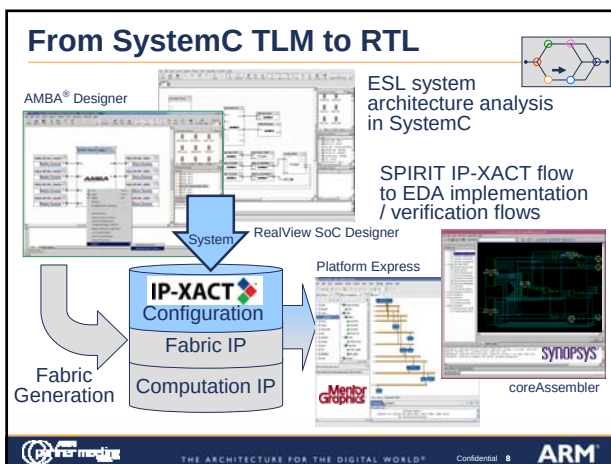
The Product Design Flow

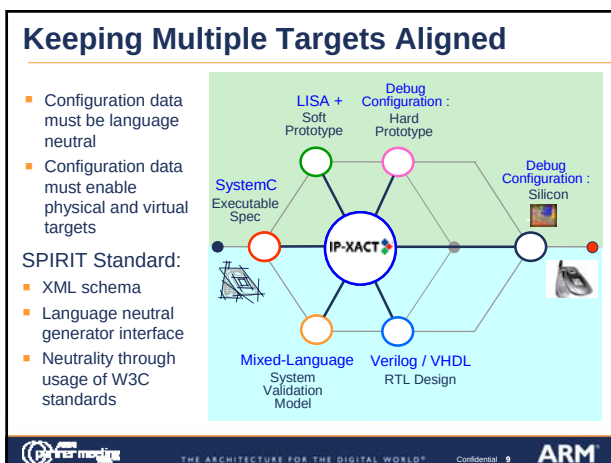


Tools for the Entire Design Flow









SPIRIT IP-XACT Programs in ARM

- IP-XACT support in the RealView SoC Designer tool
 - v6.1 SP1 IP-XACT v1.1 design-file import / export: **RELEASED**
 - IP-XACT v1.2 support: **ALPHA NOW, PRODUCTION IN THE SoC DESIGNER v7.0 TOOL**
- IP-XACT support in the AMBA Designer tool
 - AMBA busDefs for IP-XACT: **RELEASED (www.arm.com)**
 - PL300 RTL generation with IP-XACT: **RELEASED ON REQUEST**
 - PL301 RTL generation with IP-XACT: **Q3 2006**



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SPIRIT IP-XACT Programs in ARM

- IP-XACT IP encapsulation
 - Early release partner program on ARM1176™ subsystem: **ONGOING**
 - First production release for cores IP-XACT descriptions:
 - Phase I: ARM1176JZF-S™, ARM1136EJ-S™: **Q4 2006**
 - Phase II: ARM926EJ-S™, ARM7TDMI®: **DATES TBA (exp. Q4 06)**
 - Phase III: Cortex™-A8, ARM11™ MPCore™, ARM1156™: **DATES TBA**
- IP-XACT support for debug
 - RVI IP-XACT interpreter: **ALPHA**
 - RVD configuration from SPIRIT: **PLANNED**



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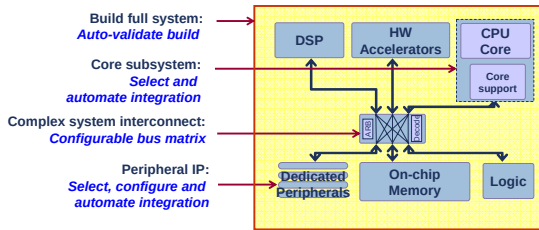


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For Advanced SoC Design Paradigms



... we need industry standards for data exchange to enable the ecosystem



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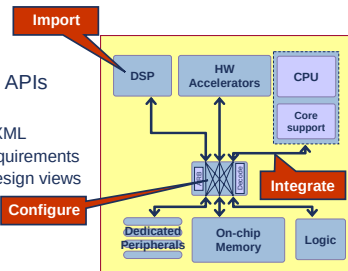
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The SPIRIT Consortium Provides...

IP-XACT

- Standard schema and APIs
- IP-XACT meta-data:
 - Machine-interpretable XML
 - Specifies integration requirements
 - Consistent across all design views
- IP-XACT generators:
 - Point-tool launch
 - IP configuration launch
 - Interface for integration with IP-XACT enabled tools

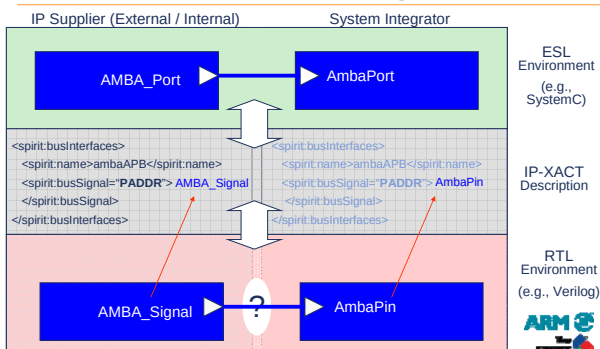


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What is an IP-XACT Description?

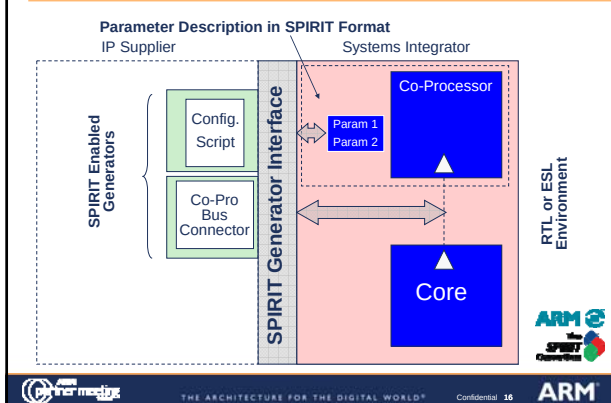


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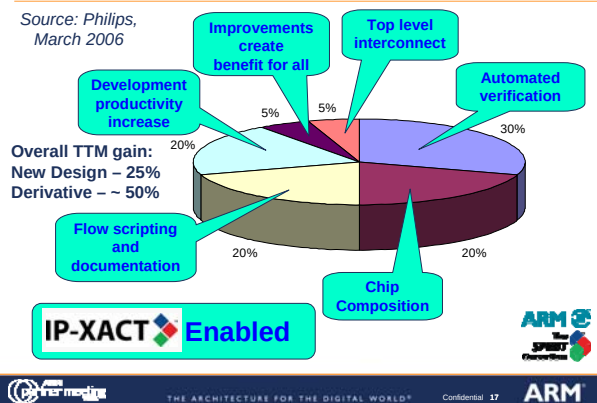
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What's an IP-XACT Enabled Generator?



Benefits from Using IP-XACT Today

Source: Philips, March 2006



The SPIRIT Consortium

- International standardization organization
 - Initially formed June 2003 (DAC Conference)
 - California non-profit organization launch: July 2006 (DAC)
- Drives deployment of it specifications
 - Each specification demonstrable at point of release
- The Consortium deliverables:
 - IP-XACT for RTL design (v1.2): Current Release
 - IP-XACT IEEE Standardization (P1685): May 06 start
 - Language-independent generator interface: Q4 2006
 - IP-EXACT ESL Extensions (for TLM): Q4 2006
- Futures:
 - IP-XACT for debug, and constraints for flow to hardware back-end

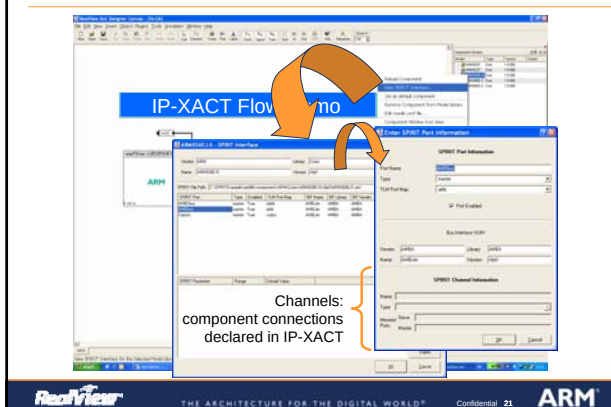
54 Members Strong, and Growing!



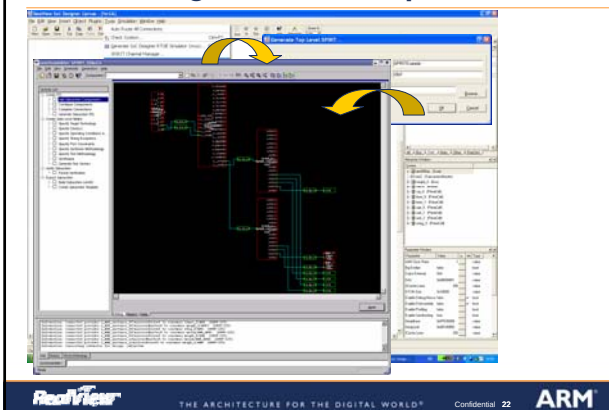
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Mapping Models to SPIRIT Information



AMBA Designer to SoC Implementation



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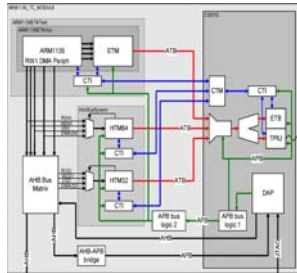
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The Problem

- Embedded systems used to be reasonably simple. E.g.:
 - 1 processor
 - 1 system bus
 - Etc.
- Telling the debugger about such systems was reasonably simple
- Now:
 - Many processors
 - Many buses
 - Complex memory maps
- So how do we debug complex systems?
 - We need the ability to debug all the processors, and to trace all the processors and buses

Debug Hardware for Complex Systems

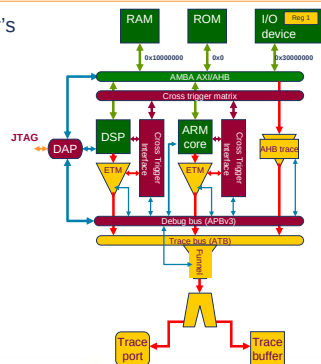
- CoreSight™ technology solves the debug hardware problem
- But there is still a software problem
 - How does the debugger know what components exist in the system?
 - How does the debugger know how the components are connected together?
 - How does the debugger know how to access the components?
 - Having lots of debug components in the system just magnifies this problem



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What Do Debuggers Need to Know?

- Description of programmer's model:
 - Programmable components
 - System buses
 - Base addresses
 - Device registers
- Debug access description:
 - Debug access to processors
 - Other debug devices
 - Debug interconnections

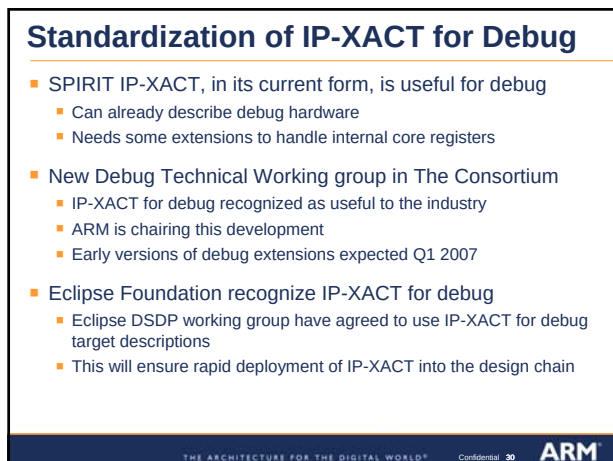
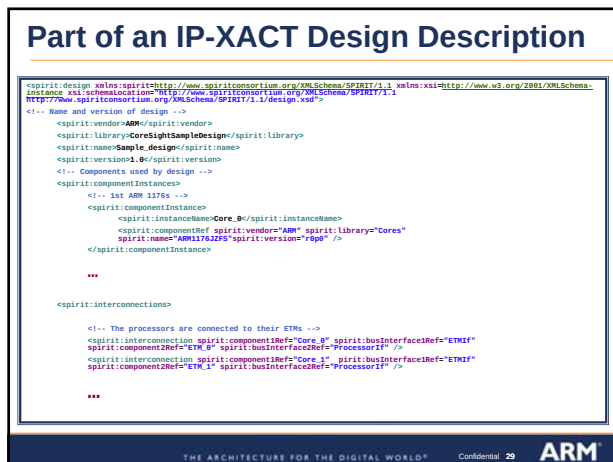
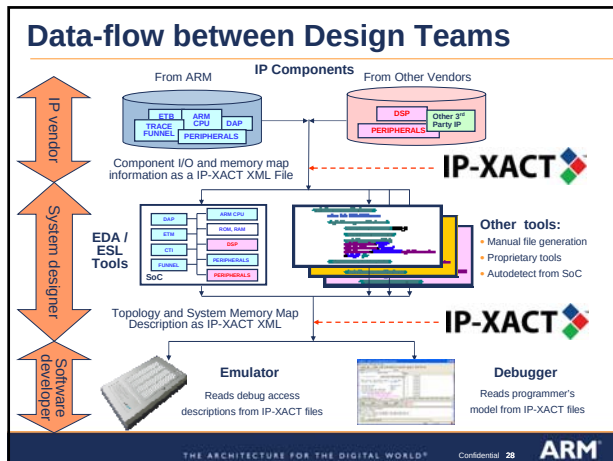


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IP-XACT Solves the Software Problem

- A SPIRIT IP-XACT system description can contain the information the debugger needs:
 - Component list
 - Component interconnection
 - Physical address maps
- Open standard, so the same descriptions can potentially be used by many tools
 - Including ARM and non-ARM debug tools
- Simple XML format
- Standard SoC design tools are starting to support the creation and use of IP-XACT descriptions:
 - Hence the IP-XACT description can be easily generated during SoC design

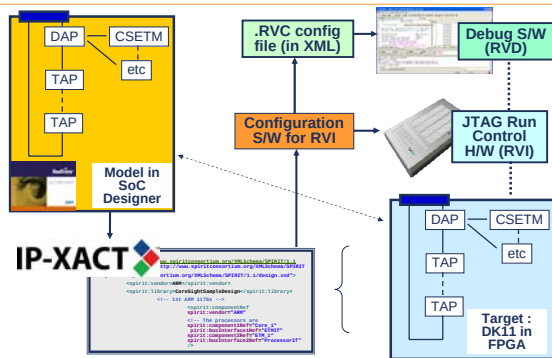
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Tools Demonstration Summary

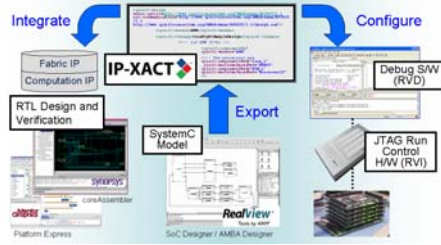


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ARM and The SPIRIT Consortium

- ARM is helping drive flow integration through The Consortium
- Debug and ESL-to-implementation flow support today
- First IP-XACT descriptions of ARM processor IP release Q4 2006



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